

Project 1: Feedback and Newcomer Retention in X Community Notes

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System

Community Notes (launched as Birdwatch in January 2021) is X's crowdsourced fact-checking program. Any enrolled contributor can attach a note to a post they believe is misleading. The note is free text, must cite a source, and is tagged by the author with a classification and reasons (such as manipulated media, outdated information, unverified claims stated as fact, missing context, satire).

Other contributors then rate submitted notes as helpful or not helpful. What makes the system unusual as a crowdsourcing mechanism is how those ratings are aggregated. Community Notes does not use a majority vote. It runs an open-source matrix factorization model that first infers a latent position for each rater from their past rating behavior, then publishes a note only if it earns approval from raters whose inferred positions differ from one another. The stated goal is to surface notes that people who usually disagree can nonetheless both accept, so agreement within a like-minded bloc, however large, is not sufficient to publish a note.

Until a note clears that bar it sits in a Needs More Ratings (NMR) state, where it is invisible to everyone except other contributors. Notes can and frequently do remain in NMR permanently. A submitted note can therefore be rated Helpful, rated Not Helpful, or remain unresolved in Needs More Ratings.

Example:

Note 1796767194967388274, written on 2024-06-01, was attached to a post showing a video of an AirDrop transfer. The author tagged it as manipulated media, argued that AirDrop has neither the behavior nor the range the video depicts, and cited Apple's own iPhone support documentation. Raters found it helpful and it was published roughly 27 hours after submission.

This is a representative successful note: short, tagged, sourced, resolved within about a day. The majority of notes do not follow this path.

Question

Does the outcome of a new contributor's first note predict whether they keep contributing, and in particular, is explicit rejection associated with different retention than receiving no final verdict?

The Wikipedia literature (e.g. Halfaker et al. on newcomer reverts; "Content is King, Leadership Lags") establishes that negative early feedback predicts newcomer departure. Community Notes is a useful contrast case for two reasons.

First, rejection here is not an editorial judgment by an established member; it is the output of a scoring rule deliberately designed to withhold approval until ideologically diverse raters converge. Failure is the expected outcome of the mechanism rather than a signal that a person disapproved of you.

Second, and more importantly, the NMR state gives us a condition that has no clean analogue on Wikipedia, where feedback is almost always explicit. We can compare contributors whose first note was explicitly rated Not Helpful with contributors whose first note never received a final verdict. Those are different experiences, and separating them is the core of what we want to measure. Our expectation going in is that receiving no final verdict may be at least as demotivating as explicit rejection, although we have no strong confidence in that prediction.

Data

X publishes the entire Community Notes corpus as free bulk downloads, with no API key, no application process, and no rate limit. We have already downloaded and inspected the two tables our question requires:

Table	Size (TSV)	Rows
notes (5 shards)	~1.5 GB	3,055,100 notes
noteStatusHistory (1 shard)	~822 MB	3,257,346 status records

`notes` gives us, per note, an author identifier, a creation timestamp, the post it was attached to, the author's classification and reason tags, and the note text.

`noteStatusHistory` gives us, per note, the field that makes this project possible: `firstNonNMRStatus` and `timestampMillisOffFirstNonNMRStatus`, recording each note's first real verdict and the moment it arrived. An empty value means the note never left NMR. Joining the two tables on note ID gives us the key explanatory variable: what happened to a contributor's first note and when.

We will group contributors into cohorts by the month of their first note and classify that note as Helpful, Not Helpful, or unresolved. We define retention as whether the contributor authors another note within 30, 60, or 90 days. Contributors whose first note is too recent for the full window are censored. Because the program grew from a small invite-only pilot to a global rollout and the scoring algorithm changed several times over five years, we will analyze cohorts separately rather than pooling across the whole period.

Two properties of the data we verified before proposing this:

1. Author identifiers are stable across data releases. We compared 40,668 notes appearing in both the 2026-09-04 and 2026-09-09 snapshots and found the author identifier matched in 100% of cases. Every longitudinal claim we want to make depends on this holding.
2. Only a rolling one-week window of snapshots is retained. Snapshot dates older than about seven days return 404 and cannot be recovered. This does not constrain the project, because each

snapshot is cumulative and contains the full history back to 2021, but it does mean we have archived a snapshot locally rather than assuming we can re-download later.

We deliberately do not use the `ratings` table, which records every individual rating and runs to tens of gigabytes, nor the text of the posts being annotated, which is not included in the public data and would require paid X API access to retrieve. Our question is answerable from `notes` and `noteStatusHistory` alone, at a scale that runs comfortably on a laptop.